AutoMetaCoder: An AI Agent for Meta-analysis Coding

Hanyi Min[1], Feng Guo[2], Sohee Kim[3], Baojiang Yang[4], James M. LeBreton[5]

[1] School of Labor and Employment Relations; Department of Psychology, University of Illinois Urbana Champaign
[2]Department of Psychology, The University of Tennessee at Chattanooga
[3]College of Business at Tennessee State University
[4]Carnegie Mellon University
[5]Department of Psychology, Pennsylvania State University

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Correspondence concerning this article should be addressed to Hanyi Min, School of Labor and Employment Relations and Department of Psychology, University of Illinois Urbana-Champaign, Champaign, IL, 61820. Email: hanyimin@illinois.edu.

Abstract

Coding in Meta-analysis is a critical component in cumulative scientific research. However, it is labor-intensive and susceptible to human error. With recent advances in large language models (LLMs), this preprint introduces **AutoMetaCoder**, an open-source AI agent designed to support meta-analytic coding through a structured, rule-governed, and human-in-the-loop workflow. AutoMetaCoder decomposes coding into modular stages, including study screening, multi-level information extraction, and standardized output organization for downstream analyses, while explicitly logging interactions between researcher inputs and model outputs to enhance transparency and auditability. We provide a detailed walkthrough of the system and present initial validation results based on published psychological meta-analyses. AutoMetaCoder achieves high retrieval and accuracy rates (97% overall) when extracting paper, sample, variable, and effect size related information across 55 primary studies with over 5,000 coding entries. These findings suggest that agentic AI workflows can substantially reduce manual coding burden while preserving methodological rigor and transparency in meta-analysis.

Keywords

meta-analysis; AI agents; large language models; evidence synthesis; reproducibility

Introduction

Meta-analysis plays a central role in cumulative science in many fields, including psychology. By aggregating finding evidence such as effect sizes across independent studies, meta-analyses enable researchers to estimate overall effects, examine sources of heterogeneity, inform decision making, and refine theories. In many domains (e.g., psychology, medicine, business, and social sciences), meta-analyses are now the primary method for adjudicating competing claims and guiding policy or clinical recommendations. As a result, the quality of meta-analysis has direct implications for scientific progress and applied outcomes, with coding remaining one of its most essential components. Coding includes determining which studies meet inclusion criteria, extracting effect size information, and recording relevant study characteristics. These coding results form the basis of all subsequent analyses and interpretations and therefore determine the overall quality of meta-analysis.

Despite its importance, meta-analytic coding remains a major methodological bottleneck. Coding large literatures is labor-intensive, difficult to scale, and vulnerable to human error and inconsistency. Researchers must make many subjective judgments, often with limited reporting detail, and even trained coders can vary in their interpretations. Coding decisions are also frequently under-documented, which makes it difficult to assess reproducibility or audit the validity of coded datasets. Together, these limitations constrain the feasibility of conducting meta-analyses across growing literatures and reduce confidence in cumulative findings (Siddaway et al., 2019).

Recent advances in large language models (LLMs) have created opportunities to automate aspects of coding process by extracting structured information from unstructured text. However, early attempts have typically relied on single pass prompting or black-box systems that

provide final outputs without transparent intermediate reasoning. Although these approaches can perform well under certain conditions, they often lack interpretability, control, and robustness. In the context of meta-analysis, such limitations are particularly problematic because coding requires careful alignment between study-specific measures and theoretical constructs, and small errors can propagate throughout the entire evidence synthesis. Moreover, naively automating coding could introduce new biases or arbitrary decisions, replacing human discretion with opaque model behavior.

In this paper, we propose a different approach. We conceptualize meta-analytic coding as a structured, rule-governed, evidence traceable workflow. Rather than using an LLM as a single pass "coder," we develop an AI agent system that breaks the process into modular stages, each with defined inputs, outputs, and researcher oversight. This enables systematic logging of decisions, traceability of extracted information back to the source text, and flexible adjustment of coding rules. We refer to this system as AutoMetaCoder, a meta-analysis agent designed to support but not replace researcher judgment. The system is intended to enhance efficiency while preserving and sometimes improving transparency by explicitly documenting the interaction between human inputs and model outputs at each stage of the workflow.

This paper (preprint preview) pursues two major objectives. First, we introduce AutoMetaCoder, an open-source meta-coding agent that integrates LLMs into a step-by-step pipeline for screening and coding, designed to support researchers by reducing repetitive manual work while producing interpretable and auditable outputs. We also demonstrate how researchers can define constructs, specify coding schemas, interact with the agent, and systematically log coding decisions. Second, we present initial validation evidence (ongoing evaluations will provide more evidence) from a benchmarking study in which AutoMetaCoder's performance is

compared with that of trained human coders using data from published meta-analyses. We evaluate the agent's ability to retrieve and extract coded variables and effect size information, and we demonstrate that AutoMetaCoder can achieve high levels of accuracy relative to human coders while offering gains in efficiency and consistency.

Collectively, this work strengthens cumulative science by demonstrating how AI can be integrated into meta-analysis workflows in a way that is scalable yet methodologically accountable, reducing burdens without sacrificing transparency, (Kepes et al., 2013; Moreau & Gamble, 2022). Meanwhile, after an explicit walkthrough of AutoMetaCoder, we also invite feedback from researchers and practitioners across related fields to inform further development and evaluation of the system.

The rest of the paper is structured as follows: First, we provide background on the role of coding in meta-analysis and clarify why it remains a persistent methodological bottleneck. We then introduce the concept of agentic flow and describe how AI agents differ from non-agentic uses of large language models, followed by a brief review of existing efforts in AI-assisted meta-analysis that motivate our system design. Next, we detail the specific steps of the AutoMetaCoder agentic workflow. Finally, we present initial validation findings and discussions in the concluding sections.

Coding in Meta-Analyses

Meta-analysis synthesizes quantitative evidence across independent studies. In its simplest form, meta-analysis involves extracting effect sizes and then combining them to estimate an overall relationship between variables or the effect of an intervention. Meta-analysis can further include moderator analyses, sensitivity analyses, and meta-regressions to examine

sources of heterogeneity, inform decision making, and refine theories. In all cases, the validity of a meta-analysis depends on the accuracy and consistency of coding decisions.

Because of its central role, coding is often one of the most time-consuming and subjective components of a meta-analysis. Meta-analysts must identify eligible studies, determine how each study operationalizes the constructs of interest, extract statistical information, and translate narrative descriptions into structured codes. These tasks require careful judgment, and they are complicated by incomplete reporting, inconsistent terminology, and variation in research designs. As a result, even well-trained coders may differ in how they interpret and code the same study.

In response, meta-analytic methodology has increasingly emphasized transparency and accountability through reporting standards (e.g., PRISMA, MARS), open-science practices, and sensitivity analyses. These efforts have improved reporting quality and encouraged the sharing of coding manuals and data. However, coding itself remains difficult to standardize and scale. Even with detailed coding protocols, substantial manual effort is required and variability across coders is often unavoidable (e.g., different coders may interpret study details differently). For large and rapidly expanding literatures, the burden of coding can limit whether meta-analyses are feasible.

Despite these advances, existing guidance and tools have only partially addressed variability in coding processes. For example, sensitivity analyses can assess how alternative inclusion criteria or analytic choices affect results, but they do not eliminate the subjectivity in coding. Similarly, open codebooks try to increase transparency but still require extensive manual effort and subjective decision-making. Instead, a system is needed that can reduce workload while making coding decisions explicit, traceable to source materials, and reproducible across analysts.

AI Agentic Flow and Its Potential in Meta-Analysis Coding

Most users interact with AI through a zero-shot question-and-answer format, in which the user asks a question and the model responds. This interaction pattern is sometimes referred to as a non-agentic workflow. In this mode, the user asks once and the model answers once, and there is limited structure for iterative reasoning, verification, or task decomposition. As a result, the model typically produces an output in a single pass, with minimal opportunity for answer checking or refinement.

The emergence of AI agents represents a paradigm shift in how language models are used. Rather than generating one-off responses, AI agents can be designed to pursue multi-step goals through structured sequences of actions. They can plan, execute, and revise actions in sequence, and they can integrate external tools (e.g., search functions, code execution, document parsing) to complete tasks. This agentic flow allows the system to break complex tasks into smaller sub-tasks, handle intermediate outputs, and maintain an organized workflow.

These features make AI agents particularly well suited for meta-analysis coding, which requires structured decision making, consistency across studies, and traceability between coded outputs and source materials. By decomposing coding workflows into modular stages–such as identifying paper relevance, searching for and extracting information from original articles, and organizing the extracted information in a structured format–agentic systems can reduce cognitive load for researchers while producing interpretable intermediate outputs.

AI agents therefore represent a promising solution for improving the efficiency, accuracy, and transparency of meta-analysis coding, an area in which structured, consistent decision making and precise construct alignment are critical. Whereas zero-shot LLMs can extract information from text efficiently, these approaches often fall short when it comes to

systematically handling the nuanced, rule-based, and error-sensitive demands of meta-analytic coding. By contrast, AI agents can more closely resemble the cognitive processes used by skilled human coders by decomposing tasks, maintaining intermediate state or memory, and adaptively adjusting coding behavior based on users' prompts.

To clarify how agentic systems achieve these advantages, we next highlight task decomposition as a core mechanism underlying agentic performance. Specifically, the agentic approach enables complex coding workflows to be broken down into smaller, well-defined sub-tasks (Wu et al., 2023; Zhou et al., 2023). Each sub-task can be addressed through an iterative process, enabling the agent to systematically revise and enhance its outputs over time. This stands in contrast to zero-shot prompting, in which the language model attempts to complete the entire meta-coding task in a single step, thereby increasing the likelihood of inaccuracies or misclassifications. Moreover, AI agents can reduce hallucination by enabling iteration and verification strategies. For example, an agent may identify uncertainty, prompt itself to gather additional evidence, or verify its output using external sources or researcher feedback. Prior work has shown that decomposing complex tasks in this way can improve the accuracy and reliability of AI systems relative to zero-shot approaches (Wu et al., 2023).

In addition, AI agents support adaptive problem solving (Yao et al., 2022), which is particularly useful in meta-analysis coding, where content-area expertise is often needed to resolve ambiguities in study reporting, terminology, and construct classification. Agentic workflows can incorporate feedback loops and reflective reasoning to adaptively address such challenges. Moreover, agentic flow can be modularized, such that components (e.g., retrieval agents, summarization agents) are reusable across different workflows (Gridach et al., 2025;

Mialon et al., 2023). This modularity allows for easier adaptation to new domains, seamless integration into new pipelines, and more efficient scaling across tasks and projects.

Importantly, agentic workflows can be designed to keep researchers in the loop, allowing human oversight at key decision points and enabling adjustment of construct definitions and coding rules as needed. Rather than relying on black-box, end-to-end automation, agentic systems produce intermediate outputs that can be reviewed, audited, and used to improve coding guidelines over time. This human–AI collaboration is especially important for meta-analysis, where accurate coding depends on careful alignment between study-specific measures and theoretically defined constructs, as is common in psychological research.

Finally, agentic systems can log prompts, intermediate outputs, and subjective decision-making details, ensuring that one of the core challenges of meta-analysis–transparency and replicability – is explicitly addressed. As a result, AI agents not only accelerate and automate repetitive or tedious coding tasks, but also make explicit the reasoning processes that domain experts use when coding primary studies.

LLM and Agent Assisted Evidence Synthesis

A growing body of work has begun to integrate LLMs into systematic review and meta-analysis workflows. Existing efforts vary widely in scope, ranging from LLM-assisted tools that target individual stages of the review process to more comprehensive review pipelines.

Several recent studies have explored using LLMs to support specific steps of evidence synthesis. For example, Ahad et al. (2024) demonstrate how LLMs can be used to assist with summarization and synthesis of study abstracts, and Khan et al. (2025) introduce an LLM-based tool that generates narrative reports from completed meta-analytic results. On the other hand, more recent work adopts explicitly agentic designs for evidence synthesis. One example is the

study by Cao et al. (2025), which presents an agent-based system for literature screening and data extraction that benchmarks performance across multiple published systematic reviews. Their study demonstrates that agentic workflows can achieve high sensitivity while maintaining strong specificity. Beyond screening and extraction, several frameworks aim to automate larger portions of the review pipeline using multi-agent architectures. Rouzrokh et al. (2025) introduce LatteReview, a modular multi-agent framework that supports title and abstract screening, relevance scoring, and structured data extraction. Xu et al. (2025) propose Manalyzer, a multi-agent system designed to support end-to-end review workflows, including search, screening, extraction, analysis, and report generation, where agentic design is perceived as a strategy for mitigating hallucinations and managing complexity. Together, these systems illustrate how agentic workflows can be scaled and adapted across different review tasks and domains.

Despite these advances, most existing tools either focus on isolated stages of evidence synthesis or prioritize end-to-end automation without explicitly treating meta-analysis coding as a structured, rule-governed process. In particular, systems designed to support features such as construct alignment, systematic logging of coding decisions, or transparent linkage between coded outputs and source text are lacking. These limitations motivate the development of AutoMetaCoder, which treats meta-analytic coding as an agentic, step-by-step workflow that emphasizes transparency, auditability, and researcher control. We will release AutoMetaCoder as an open-source system, allowing for the rigorous validation and extension of our workflows, thereby fostering cumulative and reproducible scientific progress.

AutoMetaCoder AI System Overview

The AI agent described in this study, AutoMetaCoder, is an end-to-end system for supporting the coding of meta-analysis. Designed to handle the full pipeline, from processing

PDF documents to the output of structured coded information, AutoMetaCoder supports researchers in synthesizing large bodies of literature with increased efficiency, scalability, and consistency. The pipeline begins with the input of full-text research articles or selected document sections and proceeds through several structured stages, including relevance screening, multi-level information extraction (e.g., study, sample, variable, and effect size), and the synthesis of outputs into standardized formats suitable for downstream statistical analysis.

An important feature of AutoMetaCoder is its modular integration of multiple large language models (LLMs), each tailored to specific stages of the meta-analytic workflow. Rather than relying on a single, general-purpose model, the system adopts a distributed design in which different models are invoked based on the nature of the subtask. For example, GPT4.0 model is employed to interpret researcher-provided inputs (e.g., inclusion criteria, construct definitions, and coding guidelines) thereby aligning the system's behavior with the theoretical and methodological aims of the meta-analysis. A key strength of this design is its user-configurability: researchers and developers are free to select the LLM most appropriate for each stage of the pipeline. For instance, users may choose between GPT-4.0, GPT-5.0, or other open-source or proprietary models depending on performance needs, cost constraints, or domain alignment. Similarly, different parsing engines can be substituted based on the complexity of the PDF inputs or formatting requirements. This modularity also ensures that the system remains adaptable to rapid advancements in model development; when a newer version of an LLM becomes available, it can be seamlessly integrated into the corresponding module without disrupting the overall workflow (Wu et al., 2023; Mialon et al., 2023).

To ensure methodological rigor and domain relevance, AutoMetaCoder adopts a human-in-the-loop design in which researchers play the leading role in defining and guiding the coding

process. Researchers provide content-area expertise through structured prompts, such as specifying construct definitions, identifying relevant measurement instruments, and clarifying potential coding complexities (e.g., distinctions between state- and trait-level constructs). Within this framework, the AI agent functions as a standardized automation tool that executes researcher-defined rules and workflows at scale. This hybrid approach allows the agent to scale the coding process across a large corpus and still benefit from the nuanced judgments and contextual expertise that human researchers provide. By combining the power of large language models with domain-specific workflows and expert oversight, this AI agent significantly reduces the time and labor associated with meta-analytic coding, while maintaining transparency, replicability, and methodological integrity.

Specific Steps of the Agentic Flow

AutoMetaCoder operates through a structured, multi-stage pipeline designed to support automated document screening and meta-analytic data extraction. The workflow proceeds through the following stages.

Step 1: Document parsing and processing. The AutoMetaCoder starts with loading and parsing the full-text PDF of a research article. This step involves extracting raw text and organizing document content (e.g., headings, tables, paragraphs) into a structured format suitable for downstream tasks. The current version of AutoMetaCoder employs LlamaParse (LlamaIndex, 2024) during the document processing phase to achieve high-fidelity parsing of full-text PDFs, including the accurate extraction of both narrative text and tabular data.

Step 2: Evaluating paper relevance. The parsed content is evaluated to determine the relevance of the paper with respect to the inclusion criteria of the meta-analysis pre-defined by the researcher. If the document fails to satisfy inclusion criteria, it is deemed irrelevant and the

workflow terminates at this stage. Otherwise, the agent proceeds with detailed information extraction.

Step 3: Searching and extracting information. Following the relevance assessment, AutoMetaCoder initiates a multi-layered information searching and extraction process comprising several sequential subcomponents. First, AutoMetaCoder extracts study-level metadata, including paper title, journal title, and publication year. Second, AutoMetaCoder extracts sample-level characteristics, such as the number of independent samples reported, sample sizes, and demographic attributes (e.g., age, percentage of sex, and geographic region). Third, the agent identifies and extracts variable-level information within each sample. This includes the psychological constructs under investigation, measurement instruments used, and relevant statistical descriptors such as means, standard deviations, and scale reliability. Finally, the agent extracts effect size metrics required for meta-analysis, such as Pearson's *r*, Cohen's *d*, or other effect sizes. This structured, hierarchical extraction pipeline ensures comprehensive and standardized data capture across all relevant dimensions of each study.

For meta-analysis coding, information accuracy is critical for scientific rigor. We adopted retrieval augmented generation (RAG) in this step. RAG is an advanced framework that retrieves relevant information from original sources and uses that information when generating output, grounding its responses in concrete evidence (Lewis et al., 2020). RAG-based systems retrieve relevant documents or passages from a corpus (i.e., one specific article to be coded) and condition the outputs on this retrieved context. This integration mitigates one of the key limitations of standard LLMs, hallucination (defined as the generation of factually incorrect or unverifiable content), by anchoring responses in verifiable, external sources (Lewis et al., 2020; Shuster et al., 2021). In the context of information extraction, particularly for applications

requiring high factual fidelity such as systematic reviews, meta-analyses, and scientific summarization, RAG architectures have been shown to improve both precision and interpretability (Lewis et al., 2020; Izacard & Grave, 2021). For example, when coding an effect size, the system retrieves the regression table and extracts the reported coefficient and standard error for the focal predictor. By restricting extraction explicitly stated in the article, the system avoids inferring or fabricating statistics that are not reported.

Step 4: Organizing information. After completing the data extraction process, AutoMetaCoder organizes the extracted information into a standardized format suitable for downstream meta-analytic procedures. Specifically, study-level, sample-level, variable-level, and effect size-level data are each compiled into separate spreadsheets and linked using unique identifiers. Once the organization is complete, or if the document was previously determined to be irrelevant, the agent exits the workflow for that paper, enabling the system to proceed with the next document in the corpus.

In addition to the core system, a web-based platform for AutoMetaCoder has been developed and will be released alongside the open-source code. Table 1 summarizes the key actions taken by researchers when using AutoMetaCoder, and Figure 1 illustrates the researcher-led, human-in-the-loop workflow. Figure 2 provides a screenshot of the current web-based platform.

Table 1. Step-by-Step Workflow for Using the AutoMetaCoder Platform

Step	Researcher Action
1. Define key constructs	Identify and operationalize the independent and dependent variables for your meta-analysis.
2. Conduct literature search	Locate and download relevant primary studies as PDF files. During this process, document how each variable is labeled and measured in each study, particularly when the focal constructs are unfamiliar or broadly defined. Revise your definitions of key constructs if necessary.

3. Upload primary studies	Upload the PDF files to the web-based AutoMetaCoder platform. Use the "Browse files" button on the left-side panel to select and upload one or more PDFs from your computer. You can upload multiple files at once if they are stored in the same folder, or upload files from different folders in separate actions. While the platform sets no limit for papers to be processed at once, it is recommended to start with no more than 5 papers initially to ensure the process works smoothly.
4. Specify variable names	Input the names of your independent and dependent variables in the designated text boxes beneath the file upload section. You can enter multiple variables, separated by commas.
5. Provide additional prompts	Click the "Advance" button to reveal four optional text fields. These allow you to provide detailed contextual instructions that improve extraction accuracy.
5.1. Paper relevance instructions	In the first prompt box, specify criteria for determining whether a paper is relevant to your meta-analysis. You can customize or modify the default instructions, which typically include: *1. Include papers with both variables of interest. 2. Include papers with conceptually related variables. 3. Include papers even if both variables appear as independent variables.*
5.2. Paper metadata instructions	Use paper meta information instructions field to guide how the AutoMetaCoder extracts metadata about each paper. For example: "If the metadata is presented in multiple languages, prioritize English."
5.3. Sample extraction instructions	Provide instructions for identifying and coding samples within each paper. The default prompt is: *1. Identify the number of samples in the paper. 2. For each sample, provide a name and description, then extract sample characteristics. 3. Leave fields blank if information is unclear; do not guess.* You can also give paper-specific guidance (e.g., "If Study 1 and Study 2 samples are combined, treat them as one sample.")
5.4. Variable extraction instructions	Use this field to define focal constructs and list relevant variable names and measurement scales. For example: "Affective well-being is defined by momentary and prolonged mood or affective valence. The dependent variable, affective well-being, can include the followings: positive affect, negative affect, emotion, mood, sadness, happiness, subjective well-being, psychological well-being, distress, depression, stress, and emotional state. The scales for measuring affective well-being include but not limited to BDI, DASS, PANAS." It can also be customized for a specific paper, such as "Treat Depression (BDI) as a dependent variable under the construct of affective well-being." Providing specific scale names and variable examples helps AutoMetaCoder identify relevant content.

6. Process papers	Click the "Process papers" button. Processing time will vary depending on the number and length of the PDFs. A progress bar will appear on the right. Once complete, a message will confirm successful analysis and results will display.
7. Review and download outputs	The results panel contains four tabs: **Papers**, **Samples**, **Variables**, and **Correlations**. - **Papers**: title, journal, year, type, and relevance - **Samples**: sample size, recruitment, age, gender, ethnicity - **Variables**: variable names, scales, reliability (Cronbach's alpha), descriptive statistics - **Correlations**: correlations among variables and type of association. You can view results on the webpage or download them as individual CSVs or a ZIP archive.
8. Review results and revise prompts	Carefully review the outputs for accuracy or study-specific irregularities. If needed, refine your prompts and repeat Steps 5–8. Iteration is recommended when new contextual features or unexpected variables emerge, as updating the customized instructions may improve the precision of subsequent outputs.

Performance Evaluation of AutoMetaCoder

Method

To evaluate the performance of AutoMetaCoder, we selected two meta-analyses recently published in Psychological Bulletin. One of these was Kam et al. (2024), which examined the relationship between mind wandering and affective well-being and was selected to evaluate the agent's ability to code correlational studies. Kam et al. (2024) included 64 primary studies. The other study is Emmers et al. (2024), which examined the relationship between discrimination and mental health, based on group differences. Because the authors of both meta-analyses made available the list of included studies and the coded meta-analytic data (e.g., sample sizes and effect sizes), a member of our author team identified all primary studies included in the meta-analysis, coded them using AutoMetaCoder, and compared the resulting outputs with the published coding manually. When discrepancies were observed between AutoMetaCoder's outputs and the published datasets, these cases were carefully reviewed by the authors. The

details are not discussed here. At this stage, the initial evaluation comprises 55 primary studies from Kam et al. (2024).

Metrics

The meta-analysis agent produces four separate output files for each meta-analysis: paper-level information, sample characteristics, variable coding, and correlational effect sizes. For each output file, the agent's extracted information was evaluated against the published coding and classified as correct (i.e., matching the published coding), missing (i.e., not extracted), or incorrect (i.e., extracted but not matching the published coding). Accuracy rates were calculated separately for each output file.

Results

Overall, for the initial evaluation, the AI-powered meta-analysis agent demonstrated a high level of accuracy in extracting information from 55 primary studies (See Table 2 for breakdown details). For the paper information file, 100% of entries were retrieved/extracted and all the extracted entries were coded correctly. The sample characteristics file has a retrieval rate of 100% and the retrieved entries have an accuracy rate of 98%. For the variable coding file, the agent achieved an accuracy rate of 98% (retrieve rate of 100%). Finally, for the correlational effect size file, the agent successfully retrieved 91% of the entries and 95% of the retrieved effect sizes are accurate. In summary, the agent has a retrieval rate of 97% and accuracy rate (if retrieved) of 97%.

Table 2. Initial evaluation of AutoMetaCoder

Info Type	# of Information	# of Correct (matching)	# of Missing information (not extracted)	# of Incorrect information (extracted but not matching)	% Accuracy (# of Correct /# of Total)
Paper	361	361	0	0	**100%**
Sample	790	777	3	10	**98%**
Variables	2682	2624	17	41	**98%**
Correlation	1204	1138	31	35	**95%**
Total	**5037**	**4900**	**51**	**86**	**97%**

Note. Initial evaluation includes 55 articles, with coding results generated by AutoMetaCoder on the first attempt.

Discussion

The present work introduces AutoMetaCoder, an AI agent designed to support meta-analytic coding through a structured, agentic workflow. Initial evaluation results suggest that AutoMetaCoder performs well in extracting paper-level information, sample characteristics, variable coding, and correlational effect sizes from published primary studies. With over 5,000 entries in total, a 97% overall accuracy demonstrates that an agentic, researcher-guided approach can reliably reproduce key elements of human-coded datasets for meta-analysis. At the same time, we recognize that accuracy is critical for the rigor of scientific inference. Accordingly, AutoMetaCoder includes additional features such as confidence estimates for all extracted data. Ongoing sensitivity analyses will examine the trade-offs between accuracy and scope under various extraction settings. Note that coding results with low confidence are recommended for human review, which can support task-specific refinement and accuracy.

AutoMetaCoder is intended to serve as a supportive tool rather than a replacement for human judgment. The system is particularly well suited for reducing repetitive manual work

involved in screening studies and extracting information, while allowing researchers to retain control over construct definitions, coding rules, and final decisions. Importantly, the human-in-the-loop design encourages iterative refinement of prompts and coding instructions, where the customized instructions are logged for public reviews. Again, our motivation extends beyond efficiency and prioritizes improved transparency and replicability of decision-making throughout the meta-analysis process.

There are several limitations. First, AutoMetaCoder is demonstrated here in a psychological meta-analysis context, but the agentic workflow is broadly applicable to meta-analysis coding in other fields. Extending AutoMetaCoder to new domains is flexible and feasible. Future work should benchmark performance across disciplines as well. The study screening component of AutoMetaCoder was not formally evaluated in the current benchmarking. Additional human-labeled datasets will be needed to systematically assess screening performance.

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Figure 1. Human in the loop Illustration within AutoMetaCoder Pipeline

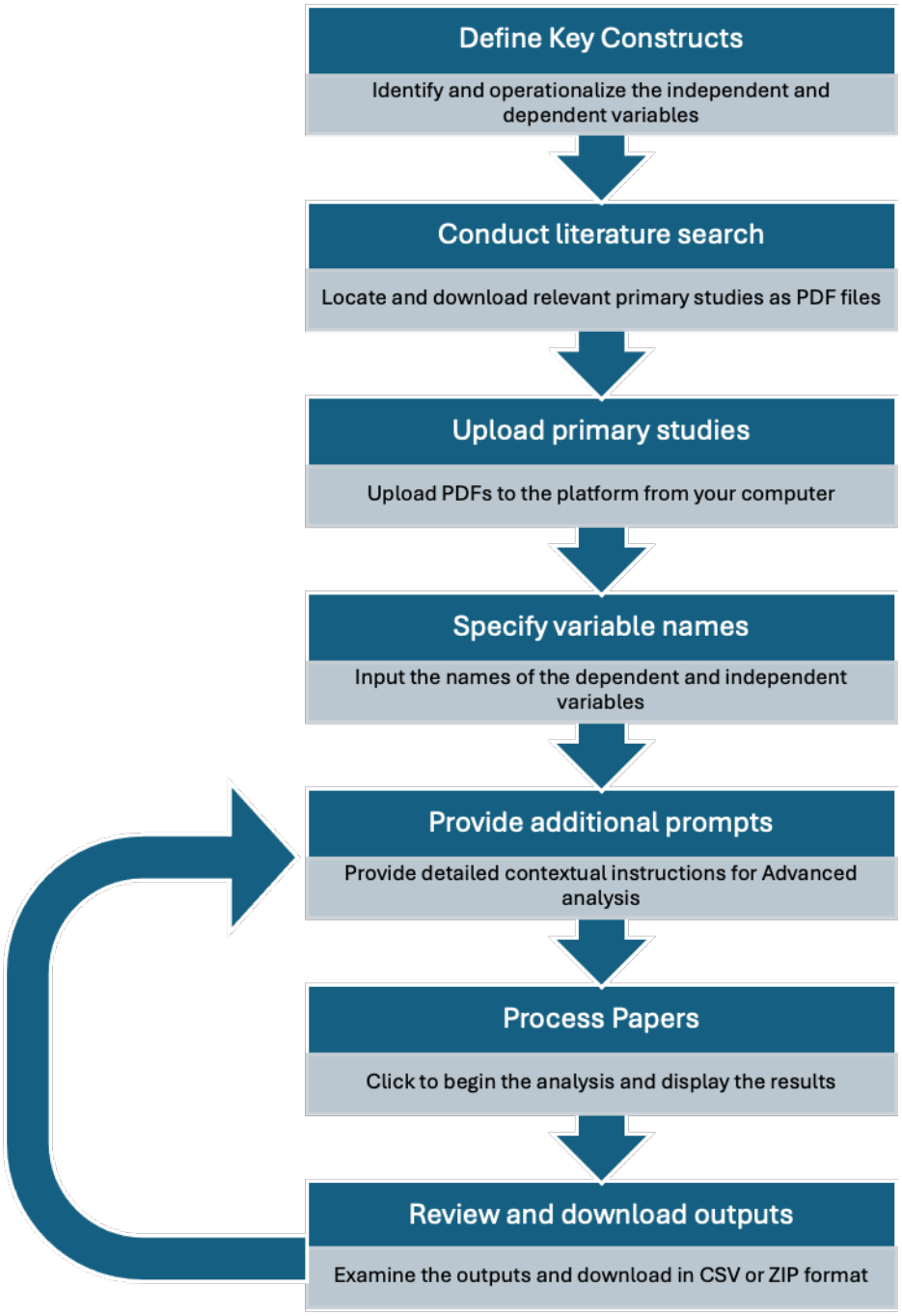


Figure 2. Screenshot of Web-based AutoMetaCoder Platform.

